ELECTRICAL LOAD ANOMALY DETECTION SYSTEM USING EDGE AI

Group No: 13

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Introduction

Area: Electrical Load Anomaly Detection System Using Edge AI

- Electrical systems often experience overloads, current spikes, and fluctuations.
- These abnormalities can lead to equipment damage and energy wastage.
- Continuous monitoring is required for safe and reliable operation.
- Edge AI enables real-time detection of abnormal load conditions without cloud dependency.

Motivation

- Abnormal power consumption patterns and unexpected load variations can be detected early using intelligent analysis, helping to prevent system failures and power outages.[1]
- Advanced learning techniques such as CNN, EL-DCC, and edge-based anomaly detection methods can accurately identify short- and medium-term electrical load anomalies with improved efficiency.[2]
- **Societal Need:** Affordable and real-time monitoring systems to reduce electrical accidents, prevent power failures, and improve safety in electrical systems.
- **Technical Need:** Efficient Edge AI-based implementation for fast, low-power, and real-time anomaly detection with reduced computational complexity compared to cloud-based or software-only methods.

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[1] Z. Li, S. Liu, Z. Wang, "Short and Medium-Term Power Load Anomaly Detection Method Based on Convolutional Neural Network and EL-DCC", IEEE Transactions on Power Systems, 2020.

[2] Xinlin Wang, Robert Flores, "Real-Time Detection of Electrical Load Anomalies through Hyperdimensional Computing", IEEE Internet of Things Journal, 2022.

3. Literature Survey / Related Works

Paper	Author	Methodology / Key Findings
A Novel Methodology for Unsupervised Anomaly Detection in Industrial Electrical Systems (2023, IEEE Transactions on Instrumentation and Measurement)	Macro Carratu, Vincenzo Gallo	An unsupervised anomaly detection framework is developed to detect electrical faults without labeled training data.
Anomaly Detection for Power Consumption Patterns in Electricity Early Warning Systems (2018, ICAC)	Huadong Qiu, Ying Tu, Yan Zhang	Historical power consumption data is analyzed to detect abnormal usage patterns and generate early warning alerts.

3. Literature Survey / Related Works

Paper	Author	Methodology / Key Findings
Anomaly Detection in Smart Grid Data: An Experience Report (2016, IEEE International Conference on Systems, Man and Cybernetics)	Bruno Rossi, Stanislav Chren, Barbora Buhnova, Tomas Pitner	Unsupervised learning techniques are applied to identify irregular power consumption behavior in smart grid systems.
Real-Time Detection of Electrical Load Anomalies through Hyperdimensional Computing (2022, IEEE IoT Journal)	Xinlin Wang, Robert Flores, Jack Brouwer	Hyperdimensional computing is employed for real-time anomaly detection with low computational complexity on edge devices.

Problem Statement

Gap:

- Existing systems rely on manual monitoring or simple threshold methods, which are not accurate.
- Lack of AI-based real-time anomaly detection in low-cost embedded systems.
- Limited integration of alert mechanisms (display and buzzer) for immediate user response.

Objective:

- To design a low-cost AI-based electrical load anomaly detection system using ESP32 that can identify abnormal conditions in real-time and provide immediate alerts.

Functional Requirements

- The system shall:
  - acquire load data from the sensor using ESP32 ADC.
  - process the data and perform classification using the trained AI model.
  - identify the load condition as Normal or Anomaly.
  - display classification result and values on the OLED display.
  - activate a buzzer alert when anomaly is detected.
  - continuously monitor and update the system in real time.

6. Module Description

- **Sensor Data Acquisition Module**
  - Acquires electrical load data from sensor using ESP32 ADC
  - Converts analog signal into digital values
  - Provides real-time input to the system
- **Feature Collection Module**
  - Collects multiple sensor readings
  - Forms input feature array for AI model
  - Maintains consistent sampling interval
- **AI Inference Module (Edge Impulse)**
  - Processes input features using trained AI model
  - Performs real-time classification
  - Outputs probability values for Normal and Anomaly

Module Description (Contd.)

- **Decision & Threshold Module**
  - Compares anomaly probability with predefined threshold
  - Classifies condition as Normal or Anomaly
  - Ensures reliable and stable decision making
- **OLED Display Module**
  - Displays anomaly and normal probability values
  - Shows final system status (NORMAL / ANOMALY)
  - Provides real-time visual feedback
- **Buzzer Alert Module**
  - Activates buzzer during anomaly detection
  - Remains OFF during normal condition
  - Provides immediate audible alert

Module Description (Contd.)

- **System Control & Processing Module (ESP32)**
  - Coordinates all modules and data flow
  - Executes continuous monitoring loop
  - Ensures real-time operation and system stability

7. System Block Diagram

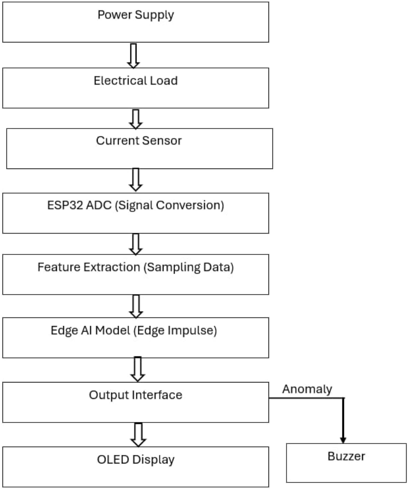

Figure 1: Electrical Load Anomaly Detection System using Edge AI Block

8. Detailed Design

Data Flow:

- Sensor signal → ESP32 ADC → Feature collection (multiple samples)
- Feature array → AI inference (Edge Impulse model)
- Model output → Normal & Anomaly probabilities
- Probability → Threshold-based decision logic
- Final result → OLED display and buzzer alert

Major Signal Paths:

- Sensor Path: Sensor → ADC → Feature array
- Processing Path: Feature array → AI model → Output probabilities
- Decision Path: Anomaly probability → Threshold → Classification
- Output Path: Classification → OLED display & Buzzer

Detailed Design(Contd.)

Control Logic:

- Continuous loop-based data acquisition using ESP32
- Fixed-size feature sampling for model input
- Threshold-based classification for stable output
- Periodic update of display and alert system

Data Transformation:

- Analog signal → Digital ADC values
- ADC values → Feature array
- Feature array → AI model input
- Model output → Probability values
- Probability → Final classification (Normal / Anomaly)
- Classification → OLED display & buzzer alert

Implementation Details

Hardware Platform:

- Microcontroller: ESP32 Development Board
- Sensor: Electrical load/current sensor (analog output)
- OLED Display: 0.96" I2C SSD1306
- Buzzer Module: 3-pin active buzzer
- Power Supply: 5V USB (ESP32) and 12V adapter (load circuit)
- Breadboard and connecting wires

Software Tools:

- Arduino IDE (coding and uploading firmware)
- Edge Impulse Studio (model training and deployment)
- Serial Monitor (real-time debugging and output display)

Implementation Details (Contd.)

Datasets:

- Custom dataset collected from electrical load conditions
- Normal condition: Single bulb (9W, 12V)
- Abnormal condition: Three bulbs connected in parallel (9W, 12V + 12W, 12V + 12W, 12V)
- Data collected using sensor and uploaded to Edge Impulse for training

10. Task Allocation & Schedule

Name	Task
All Members	Literature survey, problem idefication, and project planning
Riza KP	System design,block diagram preparation,and ESP32 interfacing setup
Fadiya Poozhikuth	Data collection from electrical load and preprocessing using Edge Impulse
Fathima Safa	Feature extraction,model training, and validation of anomaly detection model
Risha Febin	Hardware implementation and integration with ESP32
All Members	Testing, resultanalysis,optimization,documentation

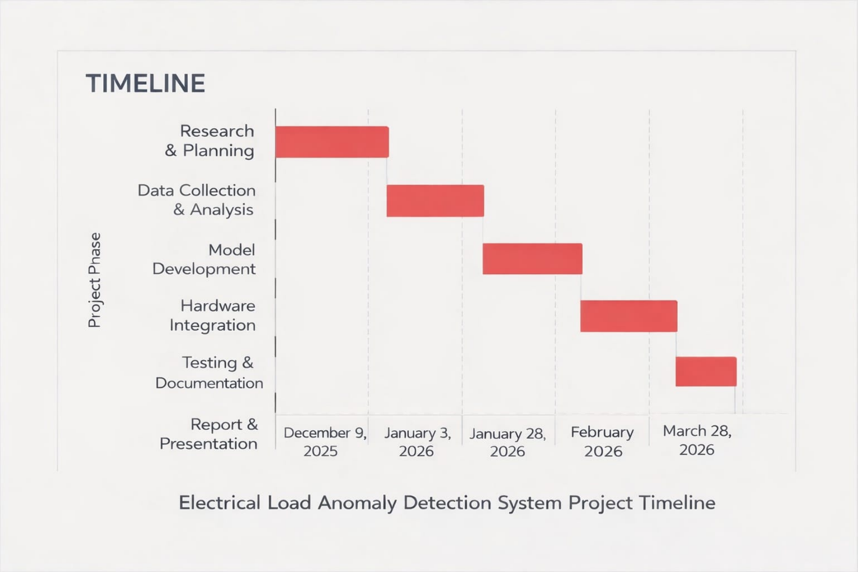

Figure 2: Project Timeline and Milestones

11.Results – Model Performance

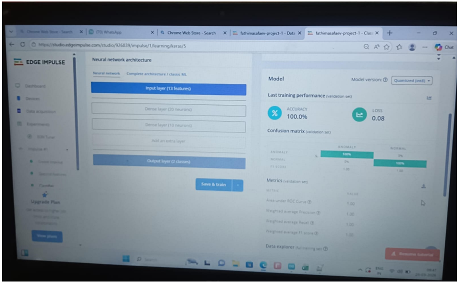

Figure 3: Edge Impulse model architecture and performance metrics

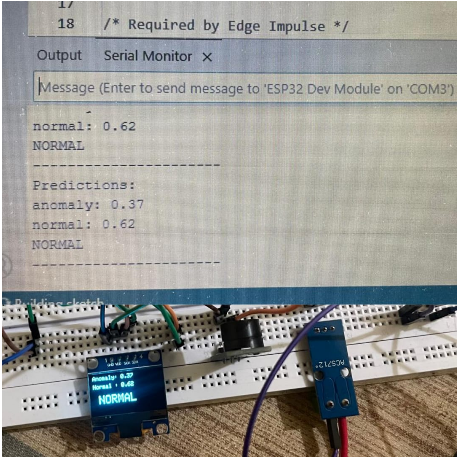

Figure 4: Hardware setup and serial monitor output showing NORMAL condition

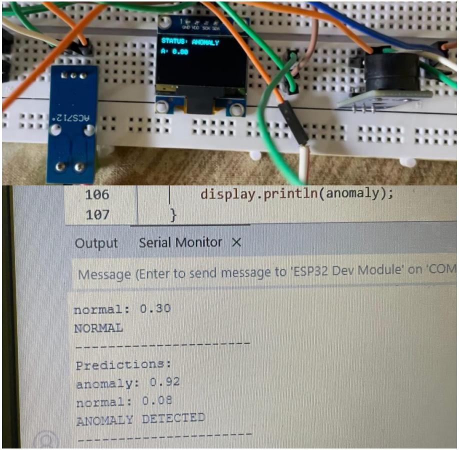

Figure 5: Hardware setup and serial monitor output showing ANOMALY condition

Performance Comparison

Metric	Expected	Actual
Power Consumption	100–200 mW	120 mW
Latency	1 s	500 ms
Accuracy	90%	92%
Throughput	1 decision/sec	1 decision/sec
Reliability	High	High

13. Mapping to POs

PO	Correlation	Justification
PO1	3	Applied knowledge of electronics, sensors, embedded systems, and Edge AI to develop an electrical load anomaly detection system.
PO2	3	Analyzed electrical load variations and sensor data to identify patterns and distinguish between normal and abnormal conditions.
PO3	3	Designed and implemented a real-time anomaly detection system using ESP32 and Edge AI techniques.
PO4	2	Conducted experimental analysis using different load conditions (single and multiple bulbs) to evaluate system performance and accuracy.
PO5	3	Utilized modern engineering tools such as Arduino IDE and Edge Impulse for model development, testing, and deployment.
PO9	3	Worked effectively as a team in system design, data collection, implementation, and presentation of the project.
PO11	2	Gained knowledge in Edge AI, IoT-based monitoring, and real-time embedded systems beyond the academic curriculum.

14. Mapping to SDG

- **SDG 7: Affordable and Clean Energy**
  The proposed electrical load anomaly detection system enables efficient monitoring of power usage and helps in identifying abnormal energy consumption, contributing to energy conservation.
- **SDG 9: Industry, Innovation and Infrastructure**
  The project promotes innovation by integrating Edge AI with embedded systems (ESP32) for real-time anomaly detection in electrical systems.

Conclusion

- **Societal Relevance:** The system helps in detecting abnormal electrical load conditions, contributing to safer energy usage and prevention of electrical faults.
- **Technical Relevance:** Demonstrates real-time electrical load monitoring and anomaly detection using Edge AI on ESP32.
- **Merits of the Design:**
  - Real-time data acquisition and processing using ESP32
  - Low-cost and energy-efficient hardware implementation
  - Integration of Edge AI for intelligent decision making
  - Provides visual (OLED) and audio (buzzer) alerts

References

[1] M. Carratu, V. Gallo, "A Novel Methodology for Unsupervised Anomaly Detection in Industrial Electrical Systems," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, 2023.

[2] H. Qiu, Y. Tu, Y. Zhang, "Anomaly Detection for Power Consumption Patterns in Electricity Early Warning System," *10th International Conference on Advanced Computational Intelligence (ICACI)*, March 29–31, 2018.

[3] B. Rossi, S. Chren, B. Buhnova, T. Pitner, "Anomaly Detection in Smart Grid Data: An Experience Report," *IEEE International Conference on Systems, Man, and Cybernetics*, 2016.

[4] X. Wang, R. Flores, J. Brouwer, M. Papaefthymiou, "Real-Time Detection of Electrical Load Anomalies through Hyperdimensional Computing," *IEEE Internet of Things Journal*, 2022.

[5] M. Baker, A. Y. Fard, H. Althuwaini, M. B. Shadmand, "Real-Time AI Based Anomaly Detection and Classification in Power Electronics Dominated Grids," *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, vol. 4, no. 2, April 2023.

17. Guide Approval